

Numericals

(b) Following training data is of a group of athletes. Based on this data, use knn algorithm and classify Sayan (Weight = 56kg., Speed=10 kmph) as a Good, Average or Poor sprinter.

Name	Weight (kg.)	Speed (kmph)	Class
Nitesh	55	9	Average
Gurpreet	58	8	Poor
Goutam	60	7.5	Poor
Gulshan	59	8.5	Average
Mohit	57	10	Good
Sahil	53	10.5	Good
Somak	53	10	Good

Eucledian dist for (56,10)

$$1.414$$

$$2.828$$

$$4.71$$

$$3.35$$

$$1$$

$$3.04$$

$$3$$

sort the distances $\rightarrow$

1, 1.41, 2.83, 3, 3.04, 3.35, 4.71

G A P G G A P

take $K=3$

$$\text{Good} = 1$$

$$\text{Avg} = 1$$

$$\text{Poor} = 1$$

(tie)

break tie using
Nearest Neighbor.

Good

$\therefore$ Sayan $\rightarrow$ Good

$$\sqrt{1^2 + 1^2} = \sqrt{2} \\ = 1.414$$

$$\sqrt{4+4} = 2\sqrt{2} \\ = 2.828$$

$$\sqrt{16+6.25} = 4.71$$

$$\sqrt{9+2.25} = 3.35$$

$$\sqrt{1+0} = 1$$

$$\sqrt{9+0.25} = 3.04$$

$$\sqrt{9+0} = 3$$

(b) Predict a class label using naïve Bayes classification, given this sample training data. The tuple you have to classify is $X = (\text{age}=\text{youth}, \text{income}=\text{medium}, \text{student}=\text{yes}, \text{credit}=\text{fair})$

Sl. No	Age	Income	Student	Credit	Buy
1	<30	High	No	Fair	No
2	<30	High	No	Excellent	No
3	31-40	High	No	Fair	Yes
4	>40	Medium	No	Fair	Yes
5	>40	Low	Yes	Fair	Yes
6	>40	Low	Yes	Excellent	No
7	31-40	Low	Yes	Excellent	Yes
8	<30	Medium	No	Fair	No
9	<30	Low	Yes	Fair	Yes
10	>40	Medium	Yes	Fair	Yes
11	<30	Medium	Yes	Excellent	Yes
12	31-40	Medium	No	Excellent	Yes
13	31-40	High	Yes	Fair	Yes
14	>40	Medium	No	Excellent	No

prior probabilities

$$\text{total} = 14$$

$$\text{Yes} = 9$$

$$\text{No} = 5$$

$$\therefore P(\text{Yes}) = \frac{9}{14}$$

$$P(\text{No}) = \frac{5}{14}$$

likelihood probability

For Yes

$$P(\text{Age}=\text{youth} | \text{Yes}) = 2/9$$

$$P(\text{Income}=\text{Medium} | \text{Yes}) = 4/9$$

$$P(\text{Student}=\text{Yes} | \text{Yes}) = 6/9$$

$$P(\text{Credit}=\text{Fair} | \text{Yes}) = 6/9$$

<30 $\rightarrow$ Youth

For No

$$P(\text{Age}=\text{youth} | \text{No}) = 3/5$$

$$P(\text{Income}=\text{Medium} | \text{No}) = 2/5$$

$$P(\text{Student}=\text{Yes} | \text{No}) = 1/5$$

$$P(\text{Credit}=\text{Fair} | \text{No}) = 2/5$$

Compute Posterior.

For Yes.

$$P(\text{Yes}|X) \propto \frac{9}{14} \times \frac{2}{9} \times \frac{4}{9} \times \frac{6}{9} \times \frac{6}{9}$$

$$= \frac{32}{547} = 0.0564 \dots$$

For No

$$P(\text{No}|X) \propto \frac{5}{14} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} \times \frac{2}{5}$$

$$= \frac{6}{875} = 0.00686 \dots$$

$$\therefore P(\text{Yes}|X) > P(\text{No}|X)$$

$\therefore$ Buy = Yes

Ans

3. Evaluate the information gain for features **CGPA** and **Communication** from the below dataset.
(5) [CO:2]

CGPA	Programming Skill	Communication	Job Offered?
High	Good	Good	Yes
Medium	Good	Good	Yes
High	Good	Bad	Yes
High	Good	Good	Yes
High	Bad	Good	Yes
Medium	Good	Good	Yes
Low	Bad	Bad	No
Low	Bad	Bad	No
Medium	Good	Bad	Yes
Medium	Bad	Good	No
Medium	Good	Bad	Yes

target classes

Yes = 8

No = 3

total = 11

$\therefore$ Entropy of Dataset (S)
(Base Entropy)

Calculate IGs

CGPA:

High ($\frac{4}{11}$)

Yes = 4
No = 0

Entropy (High)
= 0

Medium ($\frac{5}{11}$)

Yes = 4
No = 1

Entropy (Medium)
= $-\frac{4}{5} \log_2 \frac{4}{5} - \frac{1}{5} \log_2 \frac{1}{5}$
= 0.722

Low ($\frac{2}{11}$)

Yes = 0
No = 2

Entropy (Low) = 0

$$S = -\frac{8}{11} \log_2 \left(\frac{8}{11} \right) - \frac{3}{11} \log_2 \left(\frac{3}{11} \right)$$

$$= 0.845$$

$$\therefore \text{Weighted Entropy} = \frac{4}{11}(0) + \frac{5}{11}(0.722) + \frac{2}{11}(0)$$

$$= 0.328$$

$$\therefore IG(\text{CGPA}) = 0.845 - 0.328 \approx 0.517$$

Communication :

Good $\left(\frac{6}{11}\right)$

Yes = 5
No = 1

$$\begin{aligned} \text{Entropy (Good)} &= -\frac{5}{6} \log_2 \frac{5}{6} - \frac{1}{6} \log_2 \frac{1}{6} \\ &= 0.650 \end{aligned}$$

Bad $\left(\frac{5}{11}\right)$

Yes = 3
No = 2

$$\begin{aligned} \text{Entropy (Bad)} &= -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \\ &= 0.971 \end{aligned}$$

$$\therefore \text{weighted entropy} = \frac{6}{11}(0.650) + \frac{5}{11}(0.971)$$

$$= 0.796$$

$$\therefore IG(\text{Communication}) = 0.845 - 0.796 = 0.049$$

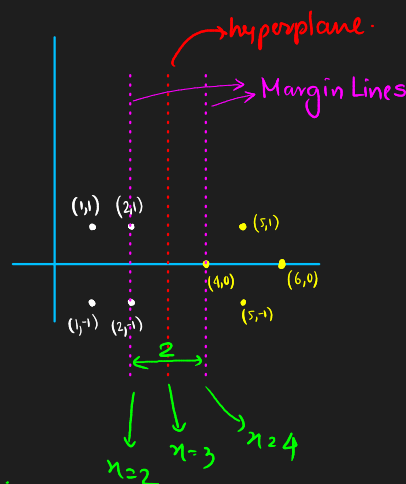
(b) Find the optimal hyperplane for the set of the data points $(1,1), (2,1), (1,-1), (2,-1), (4,0), (5,1), (5,-1), (6,0)$ and classes. Class 1: $\{(1,1), (2,1), (1,-1), (2,-1)\}$ and Class 2: $\{(4,0), (5,1), (5,-1), (6,0)\}$

find support vectors — Class 1 $\Rightarrow (2,1), (2,-1)$
Class 2 $\Rightarrow (4,0)$

The gap b/w classes is $\rightarrow 4 - 2 = 2$

Hyperplane $\Rightarrow$ The maximum margin hyperplane lies halfway between $x=2$ and $x=4$.

$$\therefore x = \frac{2+4}{2} = 3$$



1. Optimal Hyperplane $\Rightarrow x=3$ or $x-3=0$

Margin Lines $\Rightarrow x=2$ and $x=4$

Margin width $= 4 - 2 = 2 \checkmark$

Other Questions

Q. Limitations of Semi-Supervised Learning?

- Depends on data quality → if the unlabelled data contains noise or errors, model performance may decrease.
- Incorrect Assumptions → Semi-supervised algorithm assumes labelled & unlabelled data come from the same distribution.
- Error Propagation → Wrong predictions on unlabelled data may be treated as correct labels and can spread errors throughout the model.
- Complex Implementation → More difficult to design and tune than Supervised.
- Computational Cost → large amount of data can take significant time and resource.

Q. What is Seasonal index? (SI)

A seasonal index is a measure used in TSA to show the seasonal effect of a particular period (month, yr, day) compared to avg. value.

A period's value →

- Above Avg $\Rightarrow SI > 1$ (or $> 100\%$)
- Below Avg $\Rightarrow SI < 1$ (or $< 100\%$)
- Average $\Rightarrow SI = 1$ (or $= 100\%$)

$$SI = \frac{\text{Avg. value of a Season}}{\text{Overall Avg.}}$$

Suppose average ice cream sales:

Month	Average Sales
January	80
February	100
March	120

$$\text{Overall Avg} = \frac{80 + 100 + 120}{3} = 100$$

$$SI \text{ for March} = \frac{120}{100} = 1.2$$

$$\text{or } 1.2 \times 100\% = 120\% \checkmark$$

(b) Apply K-means algorithm using below given for $k=2$ (that is 2 clusters). Use $C_1(2,2)$, $C_2(1,1)$ as initial cluster centers.

Data: $A(2,2)$, $B(3,2)$, $C(1,1)$, $D(3,1)$, $E(1.5,0.5)$

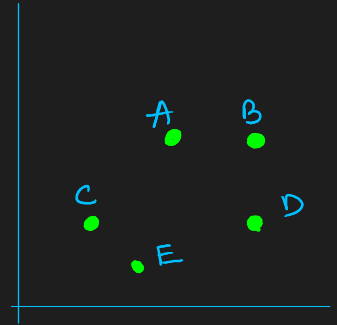
$K=2$

initial centroids $\Rightarrow C_1(2,2)$, $C_2(1,1)$

Iteration 1:

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

Pt.	$d(C_1=(2,2))$	$d(C_2=(1,1))$	Cluster
$A(2,2)$	0	1.414	C_1
$B(3,2)$	1	2.236	C_1
$C(1,1)$	1.414	0	C_2
$D(3,1)$	1.414	2	C_1
$E(1.5,0.5)$	1.581	0.707	C_2



Clusters obtained $\Rightarrow C_1 \Rightarrow \{A, B, D\}$ $C_2 \Rightarrow \{C, E\}$

Recompute New Centroids $\Rightarrow$

$$C_1 \equiv \left(\frac{2+3+3}{3}, \frac{2+2+1}{3} \right) \\ \equiv (2.67, 1.67)$$

$$C_2 \equiv \left(\frac{1+1.5}{2}, \frac{1+0.5}{2} \right) \\ \equiv (1.25, 0.75)$$

Iteration 2:

Pt.	$d(C_1=(2.67, 1.67))$	$d(C_2=(1.25, 0.75))$	Cluster
$A(2,2)$	0.75	1.46	C_1
$B(3,2)$	0.47	2.15	C_1
$C(1,1)$	1.80	0.35	C_2
$D(3,1)$	0.75	1.77	C_1
$E(1.5,0.5)$	1.65	0.35	C_2

Cluster Assignment $\Rightarrow C_1 \Rightarrow \{A, B, D\}$
 $C_2 \Rightarrow \{C, E\}$

Since the cluster assignment are unchanged from iteration 1 the algorithm converges.

Final Centroid $C_1 \equiv (2.67, 1.67)$, $C_2 \equiv (1.25, 0.75)$

7. (i) The sales of a company (in million dollars) for each year are shown in the table below

S. No.	Year	Sales
1	2005	12
2	2006	19
3	2007	29
4	2008	37
5	2009	45

Find the least square regression line, $y = ax + b$.

x	y	x^2	xy
2005	12	4020025	24060
2006	19	4024036	38114
2007	29	4028049	58203
2008	37	4032064	74296
2009	45	4036081	90405

$$\sum x = 10035$$

$$\sum y = 142$$

$$\sum x^2 = 20140255$$

$$\sum xy = 285078$$

$$n = 5$$

Now, slope

$$m = \frac{n \sum xy - \sum x \sum y}{n \sum x^2 - (\sum x)^2}$$

$$\begin{aligned} \therefore m &= \frac{5(285078) - 10035 \times 142}{5(20140255) - (10035)^2} \\ &= \frac{1425390 - 1424970}{100701275 - 100701225} \\ &= \frac{420}{50} = \boxed{8.4} \end{aligned}$$

intercept

$$b = \frac{\sum y - a \sum x}{n}$$

$$\begin{aligned} \therefore b &= \frac{142 - 8.4(10035)}{5} \\ &= \frac{142 - 84294}{5} \\ &= -\frac{84152}{5} = \boxed{-16830.4} \end{aligned}$$

$\therefore$ Regression Line $\Rightarrow y = mx + b$

$$\boxed{y = 8.4x - 16830.4}$$

Ans